CERTIFICATE

*This is to certify that Mr./Ms.*____

of __________________________________ *Class, Roll No.* _______________________

Exam No. ________________________ *has satisfactorily completed his / her term work in* ____ *for the term ending* ___________________ *in 20*___ */ 20*___.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHANGA – 388 421

Sign of the Faculty *Head of the Department*

Date:

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
Department of Civil Engineering

CONCRETE TECHNOLOGY (CL342)

INDEX

Concrete Technology (CL 342) **Date**

EXPERIMENT 1
STANDARD CONSISTENCY OF CEMENT

OBJECTIVE

Determine standard consistency of cement.

PRIOR CONCEPTS

Types of cement, chemical composition of cement, physical properties of cement

APPARTUS

Vicat apparatus with plunger and mould

Weighing balance accurate up to 0.1 gm, non-porous plate, tray, stopwatch,

Trowel, enamel tray and spatula, etc.

MATERIALS

Cement and Water

THEORETICAL BACKGROUND

The amount of mixing water affects significantly the strength and consistency of cement paste as well as that of mortar. For finding out initial setting time, final setting time, soundness of cement and strength parameters, result of standard consistency of cement test is being used. In order to compare test result with prescribed standards, it is necessary to determine the quantity of mixing water which produces a paste of standard consistency. The test is performed by using Vicat's Apparatus.

Some of the definitions of term are as below.

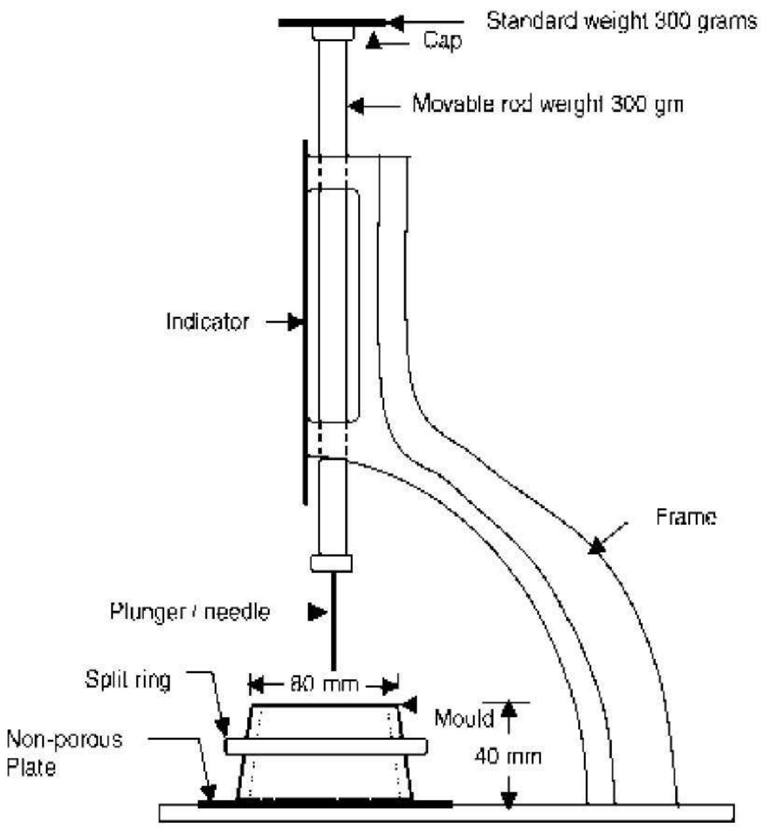
Standard / normal consistency:

It means consistency which permits the Vicat's plunger of 10mm diameter to penetrate up to a depth of 5mm to 7 mm from the bottom of Vicat's mould (gauging time 3 to 5 minutes). It is expressed as amount of water as a percentage by weight of dry cement.

Gauging time:

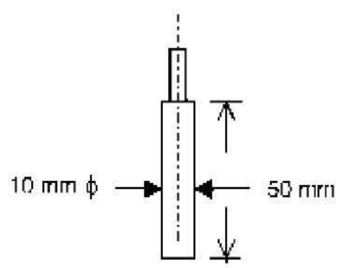
It is the period of time observed from the time water is added to cement for making cement paste till commencing the filling of mould of Vicat's apparatus in this test.

DIAGRAM



Proposition 2 : Vicat's Plunger

Plunger used to determine the standard consistency of cement is a special plunger used with Vicat apparatus. having 10-mm dia. and 50 mm length.



STEPWISE PROCEDURE (as per I.S : 4031-1988)

1. Weigh about 400 gm of cement accurately and placed it in enamel tray.
2. To start with, add about 25% of potable water and mix it by means of spatula. Care should be taken that the gauging time is not less than three minutes and not more than 5 minutes.
3. Apply thin layer of oil to inner surface of mould. Fill the Vicat's mould with this paste in the mould resting on non- porous plate.
4. Make the surface of cement paste in level with the top of mould with the trowel. The mould should be slightly shaken to the expel air.
5. Place the mould together with the non-porous plate under the rod bearing the plunger so that it touches the surface of the test block.
6. Release quickly the plunger allowing it to sink in the cement paste in the mould. Note down the penetration of the plunger in the cement paste, when the penetration of plunger becomes stable in the mould.
7. If the penetration of plunger in the cement paste is less than the 33 to 35 mm from the top of the mould, prepare the trial paste with increasing percentage of water and repeat the above mentioned procedure until the plunger penetrate to a depth of 33 to 35 mm from top or 5 to 7 mm from the bottom of mould.
8. Expressed this amount of water as a percentage by weight of dry cement.

OBSERVATIONS

Type and brand of cement ----------------------

Grade of cement ---------------------------------

Sr.No	Quantity of cement(g) (1)	Quantity of water(ml) (2)	% of water {(2)/(1)*100}	Penetration from bottom (mm)
1.				
2.				
3.				
4.				
5.				

RESULTS

The standard consistency of cement sample is found to be -------------------%

CONCLUSION

The percentage of water required to prepare a cement paste of standard consistency is

REFERENCES

IS: 269-197 Specification for Ordinary, Rapid Hardening and low heat Portland cement

IS: 4031- 1968 Methods of physical tests for Hydraulic cement.

IS: 5513- 1969 Vicat Apparatus.

QUESTIONS

1. State the meaning of normal consistency of cement. What is the importance of this test?
2. State the meaning of gauging time and its importance. How is the rate of setting of Portland cement controlled in manufacturing process?
3. What is the effect of fineness of cement on standard consistency of cement?

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FACULTY OF TECHNOLOGY AND ENGINEERING
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Concrete Technology (CL 342) **Date**

EXPERIMENT 2
INITIAL AND FINAL SETTING TIME OF CEMENT

OBJECTIVE

To determine initial and final setting time of Ordinary Portland Cement.

PRIOR CONCEPTS

Types of cement, chemical composition of cement, physical properties of cement

APPARTUS

Vicat apparatus with Initial and final set needles, Mould, Weighing balance accurate up to 0.1gm, Non-porous tray, Stopwatch, Trowel, gloves, spatula, etc.

MATERIALS

Cement, water

THEORETICAL BACKGROUND

The term setting is used to describe the stiffness of the cement paste. Hydration of Cement which gives rise to the stiffness of the paste is a gradual process and it is therefore not possible to define exactly when the setting starts and when it get completed.

Initial setting time is the time from mixing dry cement with water till the beginning of interlocking of the gel. Initial setting time of cement is that stage in the process of hardening after which any cracks appearing do not generally reunite. Final setting time is the time from mixing dry cement with water till the end of interlocking of the gel (material develop ability to carry load).

It is very important to know the setting times. Knowing the initial setting time is important in estimating time available for mixing, transporting, placing, compaction and shaping of cement paste.

<u>Needles of penetration</u>

1. Initial setting needle, 1mm in diameter. Used in the determination of the initial setting time.
2. Final setting needle, it is a needle with a metal attachment hollowed out so as to leave a circular cutting edge 5mm in diameters and set 0.5mm behind the tip of the needle.

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About the VICAT Apparatus:

It consists of the following:

1. Ruler starting from 40 mm
(Which is height of the mould)
2. The Mould.
3. Non-porous plate.
4. The plunger.

Note: The depth is always measured from the bottom of the mould.

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STEPWISE PROCEDURE

1. Prepare a neat cement paste by gauging the cement with 0.85 times water required to give a paste of standard consistency .The stop-watch shall start at this step.
2. Fill the paste in Vicat’s mould and leveled smoothly at top. The mould is then placed below the Vicat’s needle.
3. Use the 1 mm diameter and 50 mm length needle and penetrate the sample with this needle by leaving it to free fall, and then we read the VICAT ruler scale. We do a trial each 15 minutes until the depth of penetration is 5 mm. The elapsed time from mixing the water with dry cement till this moment is called initial setting time.
4. We replace the needle with another angular one (Final setting needle) of 1 mm diameter, and penetrate the sample by it every 15 minutes till only the needle makes an impression on the paste, while the circular cutting edge of the attachment fails to do so. In other words the paste has attained such hardness that the centre needle dose not pierce through the paste more than 0.5 mm.

OBSERVATIONS

Water required for standard consistency of cement P = ----------------------

Water to be mixed with the cement for the setting time test = --------------- (ml)

OBSERVATION TABLE

Time of penetration trial (after mixing)	Penetration from top (mm)	Penetration from bottom (mm)	Finial setting time (minutes)	Setting Time in min
15 min				Initial Setting Time=
30 min				Final Setting Time =
45 min				
60 min				
75 min				

CONCLUSION

As per IS initial setting for ordinary Portland cement should be around 30(+/-) 5 minute and final setting time should be around minimum 600 (+/-) 30 minute. Here performing this experiment we obtain

Initial setting time =

Final setting time =

QUESTIONS

1. Describe the process of setting and hardening of cement.
2. What are the initial and final setting times of cement? What is their important?
3. What is rapid hardening cement? What are its constituents? Where it is suitable to use?

Grade Obtained :	Sign of faculty:	Date:

EXPERIMENT 3
SOUNDNESS OF CEMENT

SOUNDNESS OF CEMENT

OBJECTIVE

Determine of soundness of Cement using Le- Chatelier apparatus.

PRIOR CONCEPTS

Types of cement, Ingredients of cement, Normal consistency of cement.

APPARTUS

Le- Chatelier apparatus, balance accurate up to 0.1 gm, Non-porous/ glass plate, Water bath with electric heating arrangement, measuring cylinder, stop watch, trowel, tray.

MATERIAL

Cement, Water

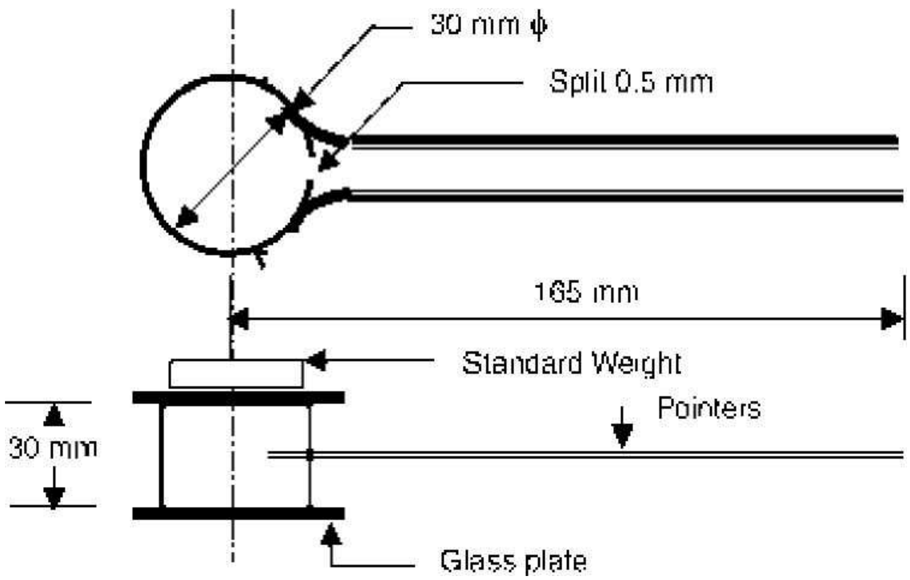
THEORETICAL BACKGROUND

It is important that cement after setting shall not undergo any appreciable change in volume. The cement is said to be sound when the percentage of free lime, magnesia and calcium sulfate are within the specified limits.

If the raw material fed into the kiln contain more lime (can combine with the acidic oxides) or if burning or cooling are unsatisfactory, the excess lime will remain in a free condition. This hard-burnt lime hydrates very slowly and, because slaked lime occupies a larger volume than the original free calcium oxide, expansion takes place. Cements which exhibit this expansion are described as unsound.

Free lime content cannot be determined by chemical analysis of the cement because it is not possible to distinguish between not reacted CaO and Ca $(OH)_2$ produced by a partial hydration of the calcium silicates when cement is exposed to the atmosphere. On the other hand, test on clinker, immediately after it has left the kiln, would show the free lime content as no hydrated cement is then present.

Dead-burnt crystalline MgO, Called periclase is deleteriously reactive, and When MgO hydration takes place, the final product is magnesium hydroxide, Mg $(OH)_2$. Because Mg $(OH)_2$ has a larger volume than its constituents, MgO concrete is expansive. However, most of the volume expansion of MgO concrete occurs at late ages (after 7 days). The expansion of MgO concrete therefore closely matches the shrinkage of mass concrete as it cools, and the concrete has been of particular interest in construction of dams to minimize crack development, simplify temperature-control measures, and speed up construction.



STEPWISE PROCEDURE (AS PER IS: 4031)

1. Prepare a cement paste formed by gauging cement with 0.78 times water required to give a paste of standard consistency. The gauging time should not be less than 3 minutes nor greater than 5 minutes.
2. Oil the inner surface of the mould. Place the mould on a glass sheet and fill it with cement paste, taking care to keep the edges of the mould gently together.
3. Cover the mould with another piece of glass sheet and peace a small weight on this covering glass sheet and immediately submerge the whole assembly in water at a temperature of 27°Cand keep it for 24 hours.
4. Take out the assembly from water after 24 hrs. Measure the distance between the indicator points and record it (D_1).
5. Submerge the mould again in water and bring the water to boiling in 25 to 30 minutes and keep it boiling for three hours.
6. Remove the mould from the water. Allow it to cool and measure the distance between the indicator points and record it (D_2).

Three samples should be tested and average of the results should be reported.

OBSERVATIONS

Types and brand of cement ……………………………………………………………..

Grade of cement …………………………………………………………………........

Weight of cement sample (Wc) =……………………gm

Water required for standard consistency (P) =……………………...%

Water added to the cement sample =0.78 x P x Wc =……………………………………..

OBSERVATION TABLE

Sr. No	Distance between pointer ends (mm)		Difference (mm)	Average (mm)
	Before Heating	After Heating		
	D_1	D_2	$D_2 - D_1$	
1.				
2.				
3.				

RESULTS

The expansion of the cement in the Le Chatelier apparatus is found to be ………….mm

BIS REQUIREMENTS

As per IS:269, when tested by Le Chatelier method, un -aerated ordinary Portland cement shall not have an expansion of more than 10 mm.

CONCLUSION

Since the average expansion of cement is………..mm and hence Cement sample is…………………… (Sound /unsound)

QUESTIONS

1. State the maximum permissible Percentage of magnesia and lime in OPC & PPC?
2. Which is another method to decide the soundness of cement?
3. Will you use unsound cement? If yes where will you use it?

4. What are the ill effects of unsound cement on structures?
5. Why the water in this test is boiled up to its boiling point?
6. State the factors which can control soundness of cement?
7. State the importance of soundness test of cement?
8. State how fineness of cement affects the soundness of cement.

Grade Obtained :	Sign of faculty:	Date:

EXPERIMENT 4
SIEVE ANALYSIS OF FINE AND COARS AGGREGATES

OBJECTIVE

Determination of fineness modulus of FA and grading of CA.

APPARATUS

A set of IS Sieves of sizes - 80mm, 63mm, 50mm, 40mm,31.5mm, 25mm, 20mm, 16mm, 12.5mm, 10mm, 6.3mm,4.75mm, 3.35mm, 2.36mm, 1.18mm, 600µm, 300µm, 150µm and 75µm.

Balance with an accuracy to measure 0.1 percent of the weight of the test sample.

MATERIALS

Fine and coarse aggregates

REFRENCES

IS: 2386(Part 1), 1963 Methods of test for aggregate for concrete.

IS: 383- 1970 Properties of natural aggregates

THEORITICAL BCKGROUND

Gradation refers as particle size distribution in aggregates. The gradation play very important role in workability and paste requirement. Gradation also control the percentage voids presents in harden concrete. The particle size distribution of a mass of aggregate should be such that smaller particle fill the voids between the larger particle. Proper grading of aggregate can produce dense concrete and needs less quantity of fine aggregates and cement paste. It is therefore essential that to produce quality concrete, the aggregate should be well graded.

TYPES OF GRADATION RELATIVELY TO THE AGGREGATE NATURE

- **Dense gradation** - A dense gradation refers to a sample that is approximately *of* equal amounts of various sizes of aggregates. By having a dense gradation, most of the air voids between the materials are filled with smaller particles. A dense gradation will result in an even curve on the gradation graph.

- **Narrow gradation** - Also known as uniform gradation, a narrow gradation is a sample that has aggregate of approximately the same size. The curvature on the gradation graph is very steep, and occupies a small range of the aggregate.
- **Gap gradation** - A gap gradation refers to a sample with very little aggregate in the medium size range. This results in only coarse and fine aggregate. The curve is horizontal in the medium size range on the gradation graph.
- **Open gradation** - An open gradation refers an aggregate ample with very little fine aggregate particles. This results in many air voids, because there are no fine particles to fill them. On the gradation graph, it appears as a curve that is horizontal in the small size range.
- **Rich gradation** - A rich gradation refers to a sample of aggregate with a high proportion of particles of small sizes

METHODS OF SIEVING

Throw-Action Sieving

Here a throwing motion acts on the sample. The vertical throwing motion is overlaid with a slight circular motion which results in distribution of the sample amount over the whole sieving surface. The particles are accelerated in the vertical direction (are thrown upwards).In the air they came out free rotations and interact with the openings in the mesh of the sieve when they fall back. If the particles are smaller than the openings, they pass through the sieve. If they are larger, they are thrown upwards again. The rotating motion while suspended increases the probability that the particles present a different Throw-action sieving orientation to the mesh when they fallback again and thus might eventually pass through the mesh.

Horizontal Sieving

In a horizontal sieve shaker the sieve stack moves in horizontal circles in a plane. Horizontal sieve shakers are preferably used for needle-shaped, flat, long or fibrous samples, as their horizontal orientation means that only a few disoriented particles enter the mesh and the sieve is not blocked so quickly. The large sieving area enables the sieving of large amounts of sample,for example as encountered in the particle-size analysis of construction materials and aggregates.

Tapping Sieving

A horizontal circular motion overlies a vertical motion which is created by a tapping impulse. These motional processes are characteristic of hand sieving and produce a higher degree of sieving for denser particles (e.g. abrasives) than throw-action sieve shakers.

Sonic sieving

The particles are lifted and forcibly dropped in a column of oscillating air at a frequency of thousands of cycles per minute. Sonic sieves are able to handle much finer dry powders than woven mesh screens.

Wet sieving

Most sieve analyses are carried out dry- But there are some applications which can only be carried out by wet sieving. This is the case when the sample which has to be analyzed is e.g. a suspension which must not be dried; or when the sample is a very fine powder which tends to agglomerate (mostly < 45 µm) - in a dry sieving, process this tendency would lead to a clogging of the sieve meshes and this would make a further sieving process impossible. A wet sieving process is set up like a dry process: the sieve stack is clamped onto the sieve shaker and the sample is placed on the top sieve. Above the top sieve a water-spray nozzle is placed which supports the sieving process additionally to the sieving motion. The rinsing is carried out until the liquid which is discharged through the receiver is clear. Sample residues on the sieves have to be dried and weighed. When it comes to wet sieving it is very important not to change to sample in its volume (no swelling, dissolving or reaction with the liquid).

LIMITATIONS OF SIEVE ANALYSIS

Sieve analysis has, in general, been used for decades to monitor material quality based on particle size. For coarse material, sizes that range down to #100 mesh (150µm), a sieve analysis and particle size distribution is accurate and consistent.

However,-for material that is finer than 100 meshes, dry sieving can be significantly less accurate) This is because the mechanical energy required making particles pass through an opening and the surface attraction effects between the particles themselves and between particles and the screen increase as the particle size decreases. (Wet sieve analysis can be utilized where the material analyzed is not affected by the liquid - except to disperse it.) suspends the particles in a suitable liquid transports fine material through the sieve much more efficiently than shaking the dry material.

Sieve analysis assumes that all particle will be round (spherical) or nearly so and will pass through the square openings when the particle diameter is less than the size of the square opening in the screen. For elongated and flat particles a sieve analysis will not yield reliable mass-based results, as the particle size reported will assume that the particles are spherical, where in fact an elongated particle might pass through the screen end-on, but would be prevented from doing so if it presented itself side-on.

Test sieving

Test sieving is a sieve analysis for determining whether the distribution of the particle sizes in a sample fulfills certain requirements.

Sieve passage

The sieve passage is the fine material that passes through the sieve during a sieve analysis.

Sieve residue

The sieve residue is the coarse material that remains on the sieve after a sieve analysis.

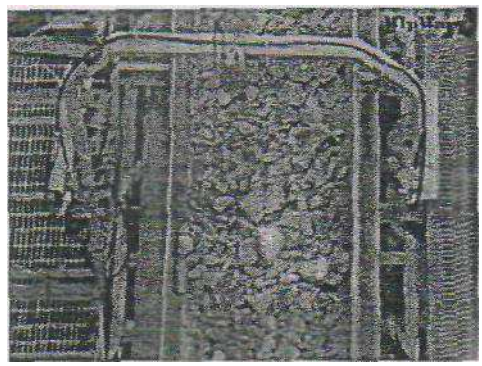
ENGINEERING APPLICATIONS

Gradation is usually specified for each engineering application it is used for. For example, foundations might only call for coarse aggregates, and therefore an open gradation is needed. Gradation is primarily a concern in pavement mix design. Concrete could call for both coarse and fine particles and a dense graded aggregate would be needed. Asphalt design also calls for a dense graded aggregate. Gradation also applies to sub grades in paving, which is the material that a road is paved on. Gradation, in this case, depends on the type of road (i.e. highway, rural, suburban) that is being paved.

Forecast

Within the last years some methods for particle size distribution measurement were developed which work by means of laser diffraction or digital image processing.

"Sieving" with Digital Image Processing



The scope of information conveyed by sieve analysis is relatively small. It does not allow for a clear statement concerning the exact size of a single particle —* it is just classified within a size range which is determined by two sieve sizes ("a particle is < than sieve size x and > than sieve size y"). And there is no additional information concerning other relevant properties like opacity or shape available. Devices which work with digital image processing enable to recall even this information and a lot more (surface analysis, etc.). The results can be fitted to sieve analysis so that a comparison between measurement results obtained with different methods is possible. Digital image processing is being used to sieve materials in mining, agriculture, and forestry industries on a regular basis.

GRADING CURVES

The curve showing the cumulative percentage of material passing the sieves represented on the ordinate with the sieve opening to the logarithmic scale represented on the abscissa is termed the grading curves. The grading curves indicate whether the grading of given sample confirms to that specified, or is too coarse or too fine, or deficient in a particular size.

1. In case the actual grading curves is lower than the specified grading curves, the aggregate is coarser and segregation of mix might takes place
2. In case the actual grading curves lies well above the specified curve the aggregate is finer and more water will be required thus, increasing the quantity of cement for constant water cement

ratio. Therefore it is uneconomical

3. If the actual grading curve is steeper than the specified it indicates an excess of middle size particle and leads to harsh mix.
4. If the actual grading curve is flatter than the specified grading curve the aggregate will be deficient in middle size particles.

TABLE 1: GRADING LIMITS FOR COARSE AGGREGATE AS PER IS:383-1970

IS Sieve designation	Percentage passing by weight					
mm	63 mm	40 mm	20 mm	16 mm	12.5 mm	10 mm
80	100					
63	85-100	100				
40	0-30	85-100	100			
20	0-5	0-20	85-100	100		
16	-	-	-	85-100	100	
12.5	-	-	-	-	85-100	100
10	0-5	0-5	0-20	0-30	0-45	85-100
4.75	-	-	0-5	0-5	0-10	0-20
2.36	-	-	-	-	-	0-5

TABLE 2: GRADING LIMITS FOR FINE AGGREGATE AS PER IS: 383-1970

IS Sieve	Percentage passing by weight			
mm	Grading Zone I	Grading Zone II	Grading Zone III	Grading Zone IV
10 mm	100	100	100	100
4.75 mm	90-100	90-100	90-100	95-100
2.36 mm	60-95	75-100	85-100	95-100
1.18mm	30-70	55-90	75-100	90-100
600 urn	15-34	35-59	60-79	80-100
300 urn	5-20	8-30	12-40	15-50
150 urn	0-10	0-10	0-10	0-15

TABLE 3: GRADING LIMITS FOR ALL- IN-ONE AGGREGATE AS PER IS: 383-1970

IS Seive designation	Percentage passing by weight of all-in-one aggregate	
	40 mm nominal size	20 mm nominal size
80 mm	100	
40 mm	95-100	100
20 mm	45-75	95-100
4.75 mm	25-45	30-50
600 um	8-30	10-35
150 um	0-5	0-6

TABLE 4: AS PER IS: 383- 1970 CLASSIFICATION OF SAND

As IS the Sand is classified in following	Fineness Modulus
Fine Sand	2.2-2.6
Medium Sand	2.6-2.9
Course Sand	2.9-3.2

PROCEDURE

1. Collect the sample and it is to be dried to a constant temperature of 110 + 5°C and weighed.
2. Arrange the IS Sieves set in descending order as per Table 1 for coarse aggregate and as per Table 2 for fine aggregate.
3. For mechanical sieving arrange set of sieve on the sieve shaker.
4. Place the sample in to top most IS sieve.
5. The sample is sieved by using a set of IS Sieves on mechanical sieve shaker.
6. On completion of sieving, the material retained on each sieve is weighed.
7. Cumulative weight passing through each sieve is calculated as a percentage of the total sample weight.
8. Fineness modulus is obtained by adding cumulative percentage of aggregates retained on each sieve and dividing the sum by 100.
9. For grading of aggregate refer figures 1,2 and 3.

OBSERVATION:

Sample weight for coarse aggregate _________________ kg

TABLE 5: SAMPLE WEIGHT FOR COARSE AGGREGATE

Sr. No	IS Sieves mm	Weight of CA retained in sieve (kg)	Cumulative of Weight retained (in kg)	% Cumulative of Weight retained	% Cumulative of Weight passing
1.	40				
2.	20				
3.	16				
4.	12.5				
5.	10				
6.	4.75				
7.	2.36				
Total					

TABLE 6: SAMPLE WEIGHT FOR FINE AGGREGATE

Sample weight for Fine aggregate _________________ kg

Sr. No	Sieves mm	Weight of FA retained in sieve (kg)	Cumulative of Weight retained (in	% Cumulative of Weight retained	% Cumulative of Weight passing
1.	10 mm				
2.	4.75 mm				
3.	2.36 mm				
4.	1.18 mm				
5.	600 u.m				
6.	300 urn				
7.	150 urn				
Total					

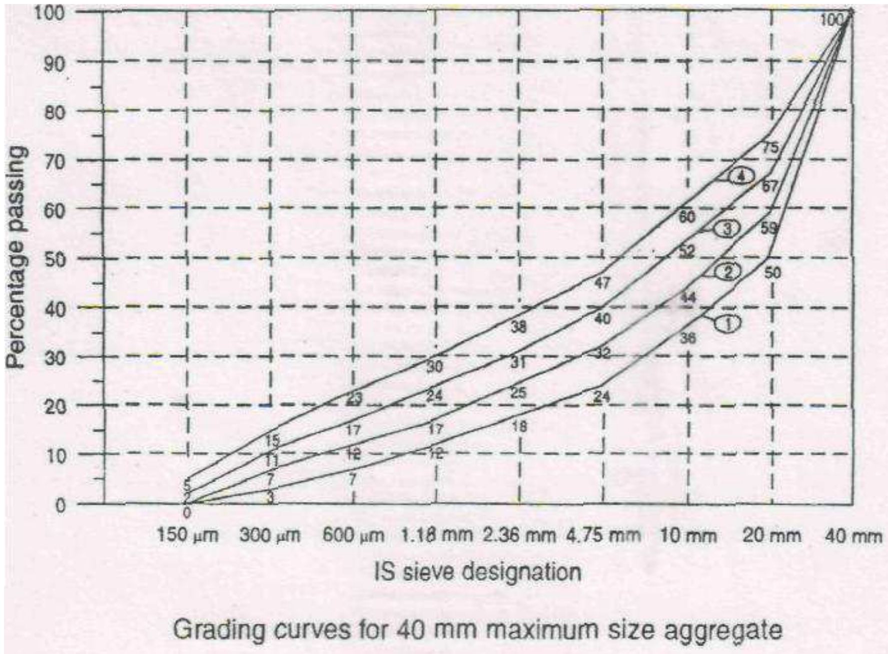


Figure 1

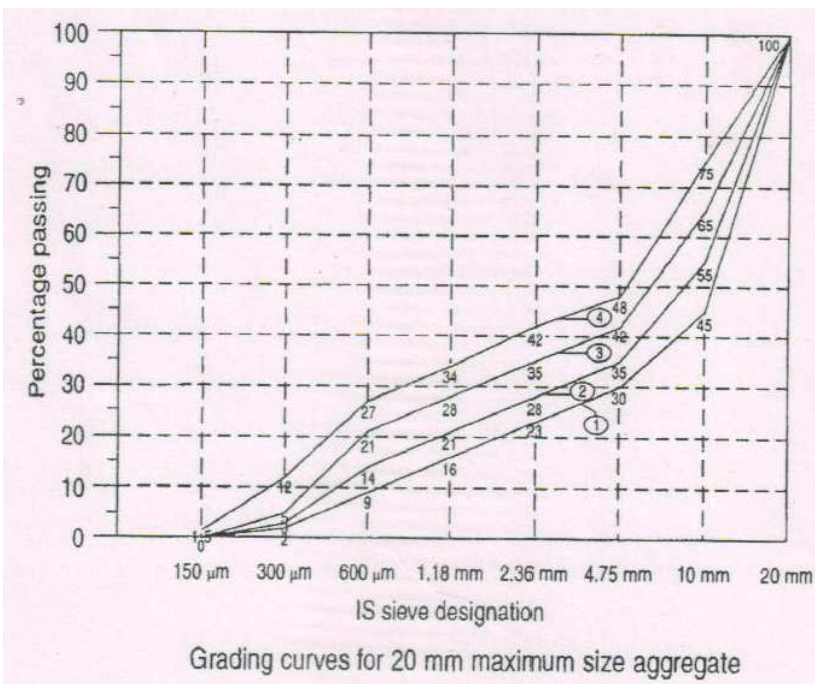


Figure 2

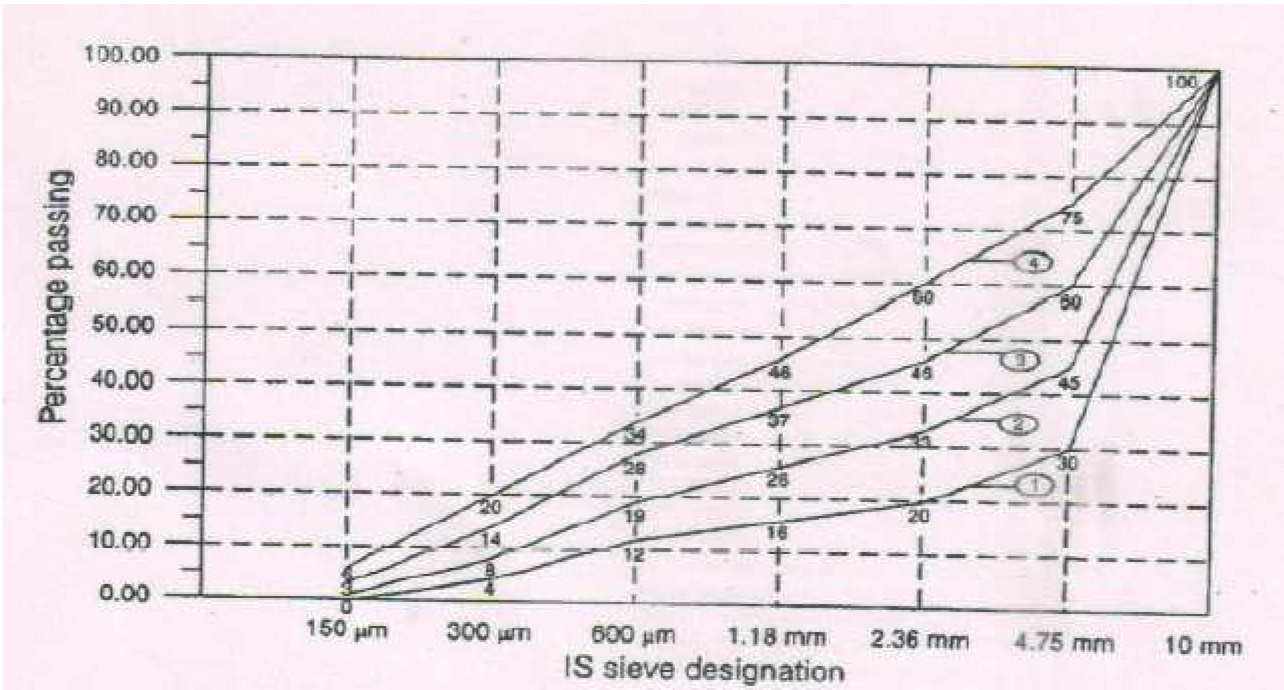


Figure 3

CONCLUSION

For Coarse aggregate

Max Size of C.A.: ___________

Shape of C.A.: _____________

Nature of Grading : ___________

For Fine aggregate

Sum of percentage of cumulative weight retained = ________

Fineness modulus = Sum of % weight retained / 100

Sand Zone: ________________

QUESTIONS

1. Why mechanical sieving is more efficient than hand sieving?
2. What is effect of grading of aggregate on workability of concrete?
3. What is effect of grading of aggregate on strength of concrete?
4. Explain all-in-one grading.
5. If F.M. of fine aggregate is 3 than give your comment about average size of particle.

Grade Obtained :	Sign of faculty:	Date:

EXPERIMENT 5
COMPRESSIVE STRENGTH OF CEMENT

OBJECTIVE

To determine compressive strength of cement.

PRIOR CONCEPTS

Types of cement, Ingredients of cement, Normal consistency of cement.

APPARTUS

Cube Vibrating machine capable of giving 1200 +/- 400 vibrations per minute, 60 tones capacity CTM, Trowel, Measuring Cylinder, Weighing balance, Non porous plate.

MATERIALS

Cement, Standard Sand, Potable Water, Oil

THEORETICAL BACKGROUND

The compressive strength of cement is an important physical property of cement. This test is a final check of the quality of cement. The test is performed as per IS: 4031(P-VI) 1988. The test is conducted on cement sand mortar cube, to avoid excessive shrinkage of cement paste. The proportion of cement and sand used for preparing the mortar is 1:3. The standard sand conforming IS 650 1966 is used for the test. The particle size distribution of sand is as per following.

Particle Size	Percent
Greater than 1 mm (Grade - I)	33.33
Smaller than 1 mm and greater than 500 microns (Grade - II)	33.33
Below500 microns (Grade - III)	33.33

The procedure for this test is described in IS: 4031 (Part- VI) 1988.

Water is to be added as (P/4+3) % of total weight of cement & sand. The size of cube mould 7.06 cm.

PROCEDURE

1. Determine the standard consistency (p) for given sample of cement.

2. Take cement 185 gm and sand total 555 gm (185 gm of each grade)
3. Mix the dry material on non porous plate.
4. Calculate water required to be added = (P/4+3) × total wt of cement & sand.
5. Add the water in to dry mix and mix thoroughly to obtain uniform colour the mixing time not more than 2 minutes.
6. Oil the inner surface of mould & place it on the vibrating table & firmly hold by clamp.
7. Fill the mold with mortar using suitably hopper 7 vibrate it for two minutes. OR fill the mortar cube in two layers and tamped 20 times by spatula to each layer.
8. Take mold from vibrating machine & finish top surface.
9. Prepare 9 cubes as per requirement and keep it in to 90% humidity 27.2^{o} C. temperature OR keep under wet gunny bags for 24 hours.
10. Remove cubes from molds after 24 hours & submerge in curing tank.
11. Take out cubes from curing tank at the time of testing, wipe the cube & place it in Universal testing machine.
12. Apply the load at the rate of 1.4 KN/cm^2/min. till the cube fails.(1.16/ KN/ sec)
13. Calculate compressive strength of concrete cube for 3 Days, 7 Days and 28 Days.

OBSERVATIONS

Type of cement ……………………

Standard consistency (P) = ……………………%

Weight of cement = …185…………………gm.

Weight of sand Grade I = …185………………….gm

Grade II = …185………………….gm

Grade III = …185…………………gm

Quantity of water {(P/4+3) percent

=……………………ml

= ……………………ml

Identification mark = ……………………………….

C/S area of cube = ………………………………

Date of casting = …………………………………

OBSERVATION TABLE

Sr. No.	Date of Testing	Age (Days)	Compressive Load P (N)	Compressive Strength $bc = \frac{P \times 10^3}{A}$ N/mm^2	Avg. Compressive Strength (N/mm^2)
1		3 Days			
2		3 Days			
3		3 Days			
4		7 Days			
5		7 Days			
6		7 Days			
7		28 Days			
8		28 Days			
9		28 Days			

RESULT

The Compressive strength for cement

3 Days = N/mm^2

7 Days = N/mm^2

28 Days = N/mm^2

Note: Draw a graph to show the development of strength gain in cement paste

IS Specification for minimum compressive strength, in N/mm^2 for different types of cement are given below.

Age	O.P.C IS: 296	P.P.C IS 1489	High Strength Cement 43 grade IS : 8112 -1989	53 Grade cement IS : 12269-1987
3 Day	16	-	23	27
7 Day	22	22	33	37
28 Day	-	31	43	53

CONCLUSION

QUESTIONS

1. What is the particle size distribution of standard sand for this test?
2. Why the standard sand is used in compressive strength test of cement?
3. Why the mould is oiled before pouring the cement mortar?
4. What is the effect of curing on compressive strength of cement?
5. Comment on the strength gain of cement

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Grade Obtained :	Sign of faculty:	Date:

Concrete Technology (CL 342) **Date**

EXPERIMENT 6
AGGREGATE CRUSHING VALUE TEST

OBJECTIVES

To determine the aggregate crushing value for course aggregate

REFERENCE

IS: 2386 – 1963 (Part IV)

APPARATUS

A 15 cm diameter open – ended street cylinder with plunger and base plate

A straight metal tamping rod of

16 mm in diameter and 45 to 60 cm long, rounded at one end.

A balance of capacity 3 kg readable and accurate to 1 gm

A compression testing machine capable of applying a load of 40 tonnes and which can be operated to give a uniform rate of loading so that the maximum load is reached in 10 minutes.

For measuring the sample, cylindrical metal measure of sufficient rigidity to retain the form under the rough usage having

- Internal Diameter - 115 gm
- Internal height - 180 mm

IS: sieves of size 12.5, 10 and 2.36 mm

Oven

MATERIAL

Coarse aggregates

THEORETICAL BACK GROUND

The crushing value of aggregate gives a relative measure of the resistance of an aggregate sample to gradually applied compressive load. Generally, this test is made on single sized aggregate passing through 12.5 mm and retained on 10 mm IS sieve. The aggregate is placed in a cylindrical mould and a load of 40 tonnes is applied through a plunger. The material crushed to finer than 2.36 mm is separated and expressed as a percentage of the original weight taken in the mould. This percentage is referred as aggregate crushing value.

The crushing value of aggregate is restricted to 30 percent of concrete used for roads and pavements and 45 percent may be permitted for other structures. When the aggregate crushing value becomes 30 or higher, the result is likely to be inaccurate in which case the aggregate should be subjected to "10 percent fines value" test which gives a better picture about the strength of such aggregates.

The standard aggregate crushing value test shall be made on aggregate passing form 12.5 mm and retained on a 10 mm IS sieve but if required, or if the standard size is not available other size up to 25 mm may be used for test but results will not be compatible with those obtained by with the standard size.

PRINCIPAL DIMENSIONS OF APPARATUS FOR AGGREGATE CRUSHING TEST

KEY TO DIMENSIONS			
Letter	Dimensions for Cylinder	150 mm cylinder	75 mm cylinder
		(mm)	(mm)
A	Internal diameter	152.0 ± 0.5	77.0 ± 0.5
B	Height	130 to 140	80 t 80
C	Wall thickness	16	16
	Plunger		
D	Diameter of piston	150 + 0.5	75.0 + 0.5
E	Diameter of stem	100 to 150	50 to 75
F	Height	100 to 115	65 to 75
G	Depth of piston	25	20
H	Diameter of hole Nominal	20	10
	Base plate		
J	Thickness (nominal)	6.3	6.3
K	Side length of square	200 to 230	110 to 115

PROCEDURE

1. For standard test, take about 3 kg material consisting of aggregates passing from 12.5 mm and retained on 10 mm IS sieve.
2. If the aggregates are not surface dry, heat it at a temperature of about 100 to 110^0C. The period of heating shall not exceed four hours.

3. Take the cylindrical metal measure of diameter 11.5 cm and depth 18cm and fill it with about aggregates in three equal layers, each layer being tamped 25 times with tamping rod of 16mm having rounded end. Finally, level the material using tamping rod as straight edge.
4. Weight the material from cylindrical metal measuring (Say W). The same weight of the sample shall be taken for repeat test.
5. The cylinder of the test apparatus shall be put in position on the base plate and the test same added in three equal layers, each layer shall be subjected be 25 strokes from tamping rod. Surface of the aggregate shall be carefully leveled and the plunger shall be placed such that it remains horizontally on this surface. Care shall be taken to ensure that the plunger does not jam in the cylinder.
6. Place the apparatus with the test sample and plunger in position, between the plates of compressor testing machine. Apply the load at uniform rate such that the total load of 40 tones is reached in 10 minutes.
7. Release the load and remove the crushed material from the cylinder.
8. Sieve the crushed material on 2-36 mm sieve.
9. Weight the fraction passing through 2.36 sieve (say W_2)
10. Calculate aggregate crushing value

$$\text{Aggregate crushing value} = \frac{\text{Weight of fraction passing through 2.36 mm ISsieve}}{\text{100\%Weight of sample taken}}$$

$$= \frac{W_2 \text{ x } 100\%}{W_1}$$

Minimum two tests shall be made and average shall be reported.

PRECAUTIONS

Avoid loss of the fine crushed particles of aggregate

See that the plunger does not jam in the cylinder.

OBSERVATION TABLE:

Sr. No.	Weight of sample taken (W_1) gm	Weight of fraction passing through 2.36 mm IS sieve (W_2) gm	Crushing value of aggregate (%) W_2/W_1 %	Average crushing value (%)
1				
2				
3				

CONCLUSION

QUESTIONS

1. What are the IS requirements for aggregate crushing value?
2. What will be your conclusion, when aggregate crushing value of sample is 50%?

Grade Obtained :	Sign of faculty:	Date:

Concrete Technology (CL 342) **Date**

EXPERIMENT 7
FLAKINESS INDEX AND ELONGATION INDEX TEST

OBJECTIVES

To determine the flakiness index and elongation index of coarse aggregate

APPARATUS

Balance and weights

Standard metal gauge

Thickness gauge

Length gauge

IS: sieves

MATERIAL

Coarse aggregates

THEORETICAL BACK GROUND

A particle of an aggregate is said to be flaky, if its least dimension is lesser than 0.6 times of its mean dimension.

Flakiness index is the weight of flaky particles measured as percentage of the total weight of the sample. Flakiness index, more than 35 to 40% is not desirable.

A particle is said to be elongated, if its length is greater than 1.8 times its mean dimension. Elongation index is the weight of elongated particles measured as percentage of the total weight of the sample. For elongation index no limit has been laid down.

The flaky or elongated particles are not desirable in concrete as their large number creates more voids, requires more fine materials and more water for the same workability. They may create inherent weakness with possibilities of breaking or rupture of aggregates due to heavy load. Such particles also reduce the strength and bond capacity and compaction of the concrete resulting very low workability of concrete.

Elongation index test and flakiness index test are not applicable to sizes smaller than 6.3 mm.

PROCEDURE:

FOR FLAKINESS INDEX

1. Take sufficient quantity of agreement to provide at least 200 pieces of any fraction to be tested
2. Sieve the sample on IS sieves as mentioned in the table
3. Try to pass aggregate particles through the corresponding slot of thickness gauge. The aggregate piece passing through 500 mm and retained on 40 mm sieve, should only be passed through 0.6 x (50 + 40) / 2 = 27.0 mm slot. If the aggregate passes through this slot, then the aggregate particle is flaky.
4. Weight all the particles which pass through this slot.
5. Calculate the flakiness index.

$$\text{Flakiness Index} = \frac{\text{Weight of material passing through the thickens gauge x 100\%}}{\text{Total weight of sample}}$$

FOR ELONGATION INDEX

1. Sieve the sample through IS sieves as specified in observation table.
2. Separate aggregate particles by passing the particles through the corresponding slot of length gauge. If the length of the particle is 1.8 (50+40) / 2 = 81 mm or more for particle passing through 50 mm sieve and retained on 40 mm sieve, it is said to have retained on the length gauge.
3. Weight all such particles
4. Calculate the elongation index as follows

$$\text{Elongation Index} = \frac{\text{Weight of material retained by the length gauge x 100\%}}{\text{Total weight of sample}}$$

PRECAUTIONS

Quantity of aggregate shall be taken sufficient to provide the minimum number of 200 particles of any fraction to be tested in each.

OBSERVATION TABLE

For flakiness index

Sr No.	Size of aggregates		Corresponding thickness gauge in mm = 0.6 x mean sieve size	Weight of sample taken in kg	Wt of material passing through T.G in kg
	Passing from IS sieve	Retained on IS sieve			
1	63 mm	50 mm	33.90		
2	50 mm	40 mm	27.00		
3	40 mm	25 mm	19.50		
4	31.5 mm	25 mm	16.95		
5	25 mm	20 mm	13.50		
6	20 mm	16 mm	10.80		
7	16 mm	12.5 mm	8.55		
8	12.5 mm	10 mm	6.75		
9	10 mm	6.3	4.89		
			Total	W_1=	W_2=

CALCULATION

$$\text{Flakiness Index} = \frac{\text{Weight of material passing through the thickens gauge x 100\%}}{\text{Total weight of sample}}$$

For elongation index

Sr No.	Size of aggregates		Corresponding elongation gauge in mm = 0.6 x mean sieve size	Weight of sample taken in kg	Weight of material passing through T.G in kg
	Passing from IS sieve	Retained on IS sieve			
1	63 mm	50 mm	33.90		
2	50 mm	40 mm	27.00		
3	40 mm	31.5 mm	19.50		
4	31.5 mm	25 mm	16.95		
5	25 mm	20 mm	13.50		
6	20 mm	16 mm	10.80		
7	16 mm	12.5 mm	8.55		
8	12.5 mm	10 mm	6.75		
9	10 mm	6.3	4.89		
			Total	W_1=	W_2=

CALCULATION

$$\text{Elongation Index} = \frac{\text{Weight of material retained by the length gauge x 100\%}}{\text{Total weight of sample}}$$

RESULT

For given sample of aggregate

Flakiness Index =

Elongation Index =

CONCLUSION

QUESTIONS

1. What is the effect of flaky or elongated particles in concrete?
2. What are the criteria for determining aggregate particle to be flaky or elongated?

REFERENCE

IS : 2363 – 1963

Grade Obtained :	**Sign of faculty:**	**Date:**

Concrete Technology (CL 342) **Date**

EXPERIMENT 8
SLUMP TEST

OBJECTIVE

To measure workability of fresh concrete by slump test

APPARATUS

Slump cone 100 mm diameters at top and 200 mm diameter at bottom and 300 mm height.

Tamping rod 16 mm and 0.6 m long with round end, Trowels, Steel scale, Measuring cylinder, Stop watch, Balance scoop, Balance, Metal tray

MATERIALS

Cement, Fine Aggregate, Coarse Aggregate 20mm size.

THEORY

Workability is defined as the property of concrete which determine the amount of useful internal work necessary to produce full compaction. In this test, mobility or flow ability of concrete without segregation is measured in quantitative terms . A workable concrete produce dense concrete with high strength and durability. The factors affecting work ability is w/c ratio, mix proportion, size, shape, texture and grading of aggregate and use of admixture. Methods of measurement of work ability is slump test Vee Bee Test, compaction factor test and flow test. Requirement of degree of workability depends upon the size of sections and compaction methods as shown in Table 1.

The slump test is very simple often used as construction sites to check batch to batch variability. Fresh concrete is filled in conical mould in four layers with standard compaction. Then mould is lifted up immediately. The subsidence is measured as slump in mm, as shown in Figure 1. Slump test is not suitable for aggregate size more than 40 mm and concrete with, very low and extremely low work ability.

PROCEDURE

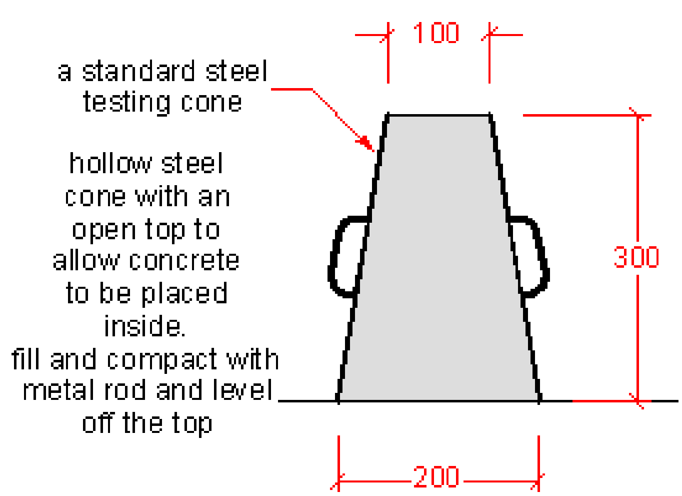
1. Oil inner surface of conical mould
2. Prepare the trial concrete in metal tray with proportion 1:2:4 w/c ratio and 0.6
3. Fill the concrete in conical mould in four equal layers (each approximately one quarter of height). Each layer tamped by 25 blows of tamping road as quick as possible and in

uniform manner over the cross section. After the top layer has been rodded, the concrete is struck off level with trowel or tamping rod.

4. The mould is lifted up by carefully in vertical direction, concrete slump.
5. The subsidence of concrete is measure by steel foot rule form height of mold to the highest point of specimen slump
6. The slump is measured in mm.
7. Take next observation as above with same proportion of concrete mix and different w/c ratio 0.65, 0.55, 0.45.

TABLE 1

	Placing condition	Degree of workability	Slum mm	Vee bee time sec	Compaction factor
1	Sections subjected to prolong vibration accompanied by pressure	Extremely low	0	Above 20	Below 0.7
2	Small sections subjected to intensive vibrations	Very low	0.25	Above 10 to 20	Above 0.7 to 0.8 to 0.85
4	Lightly reinforced section without vibration or heavy reinforced with vibration	Medium	Above 50 to 100	0 to 5	Above 0.85 to 0.95
5	Heavily reinforced section not suitable for vibration	High	Above 100	0	Above 0.95



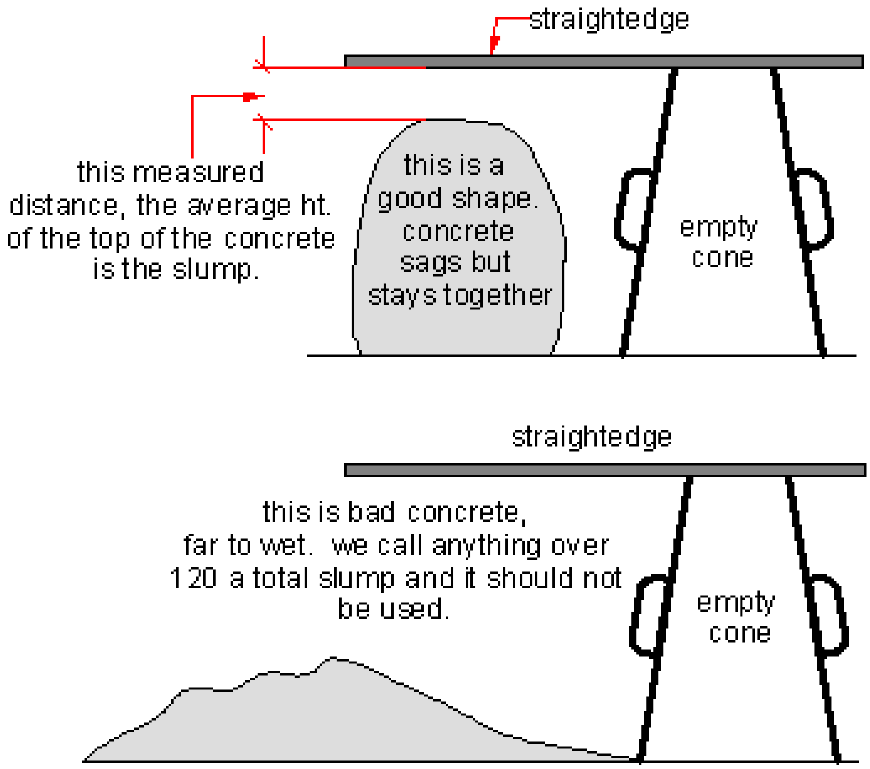


Figure 1

OBSERVATION:

Proportion of concrete mix = 1:2:4

Weight of cement = ___________ kg

Weight of fine Agg. = ___________ kg

Weight of coarse Agg. = _________ kg

Sr.No.	W/c. Ratio	Water added	Slump in mm
1	0.60		
2	0.55		
3	0.50		
4	0.45		
5	0.40		

CONCLUSION

QUESTIONS

1. Define workability of concrete.
2. List factors affecting workability of concrete.
3. What is limitation of slump test?
4. Which property mostly measures by slump test?
5. What is true slump?

Grade Obtained :	Sign of faculty:	Date:

Concrete Technology (CL 342) **Date**

EXPERIMENT 9
COMPACTION FACTOR TEST

OBJECTIVE

To measure workability of fresh concrete by compaction factor apparatus.

APPARATUS

Compaction factor apparatus, Hand scoop, Measuring cylinder, Trowels, Vibration table or tamping rod, weighing balance

THEORY

Compaction factor test give the compacting behavior fresh concrete under external force. It measures compatibility of concrete. It is very important aspect of workability. This test is most accurate than other methods. It is most suitable for any degree of workability of fresh concrete. It also measure compatibility of dry concrete mix. It is most suitable for laboratory condition. The compatibility of concrete is expressed in terms of compaction factor. It is ratio of weight of compact concrete by standard effect to the weight of fully compacted concrete. The compaction factor apparatus consist of two hoppers and a cylindrical mould in vertical line. The concrete is filled in upper hoper and allowed to fall in lower hopper then cylinder by standard effort.

PROCEDURE:-

1) Weight the empty cylinder = W1 gm.
2) Oil the inside of hopper and cylinder of compaction factor test apparatus.
3) Prepare the concrete with suitable proportions e.g 1:2:4 and w/c ratio.
4) Fill the fresh concrete in upper hopper by scoop.
5) Open the door of upper hopper, the concrete will fall in lower hopper.
6) Open the door of lower hopper concrete falls in cylinder.
7) Level the top surface of concrete in cylinder by trovel weight the cylinder with concrete = W2 gms.
8) Empty the cylinder and fill in layer of 5 cm deep and compact each layer fully.
9) Level the top surface of cylinder and weight the cylinder with concrete = W3 gm.
10) Calculate compaction factor C.F. = (W2-W1) / (W3-W1)

11) Take next three observations on of same proportion and different w/c ratio 0.65, 0.55, 0.45 respectively.

OBSERVATION

Mix proportion of concrete ____________________

Wt. of coarse aggregate ____________________ gm

Weights of cement ____________________ gm

Weight of fine aggregate ____________________ gm

Weight of empty cylinder W_1 ____________________ gm

Sr.No.	Particulars	I	II	III	IV
1	W/c ratio	0.60	0.55	0.40	0.35
2	Wt. or water added				
3	Weight of cylinder partial compacted concrete W2				
4	Weight of cylinder fully compacted concrete W3				
5	Compaction factor CF = W2-W1/W3-W1				

RESULT

Compaction factor at following W/C ratio

0.60	
0.55	
0.40	
0.35	

CONCLUSION

QUESTIONS

1. What are merits and demerits of compaction factor test?
2. What is basic principle of compaction factor test?

Grade Obtained :	Sign of faculty:	Date:

Concrete Technology (CL 342) **Date**

EXPERIMENT 10
ULTRA SONIC PULSE VELOCITY TEST

OBJECTIVE

To measure the compressive strength of concrete by Ultra sonic pulse velocity test (NDT).

APPARATUS

Ultra sonic pulse velocity test

MATERIALS

Concrete beam, cube or existing concrete element

THEORY

This test is performed to assess the quality of concrete by ultrasonic pulse velocity method as per IS:13311(Part1)-1992.

The method consists of measuring the time of travel of an ultrasonic pulse through the concrete being tested. Comparatively higher velocity (less time) is obtained when concrete quality is good in terms of density, uniformity, homogeneity etc.

UPV can be used for assessing integrity of structural concrete elements of up to 7.5 m thick with two-sided access. UPV tests can also be performed on wood, masonry, and metal structures.

<u>Basic Concept</u>: This technique involves measuring the travel time of pulses through material with a known thickness. The frequencies of the transmitted signals vary from 50 to 300 kHz. Under certain conditions, it is possible to estimate the compressive strength using the ultrasonic pulse velocity method. The method requires that objects to be tested have two sides available for access. Voids, honeycomb, cracks, delaminating and other damage to concrete can be located. This method is used to predict the strength of early stage concrete and as a relative indication of concrete quality.

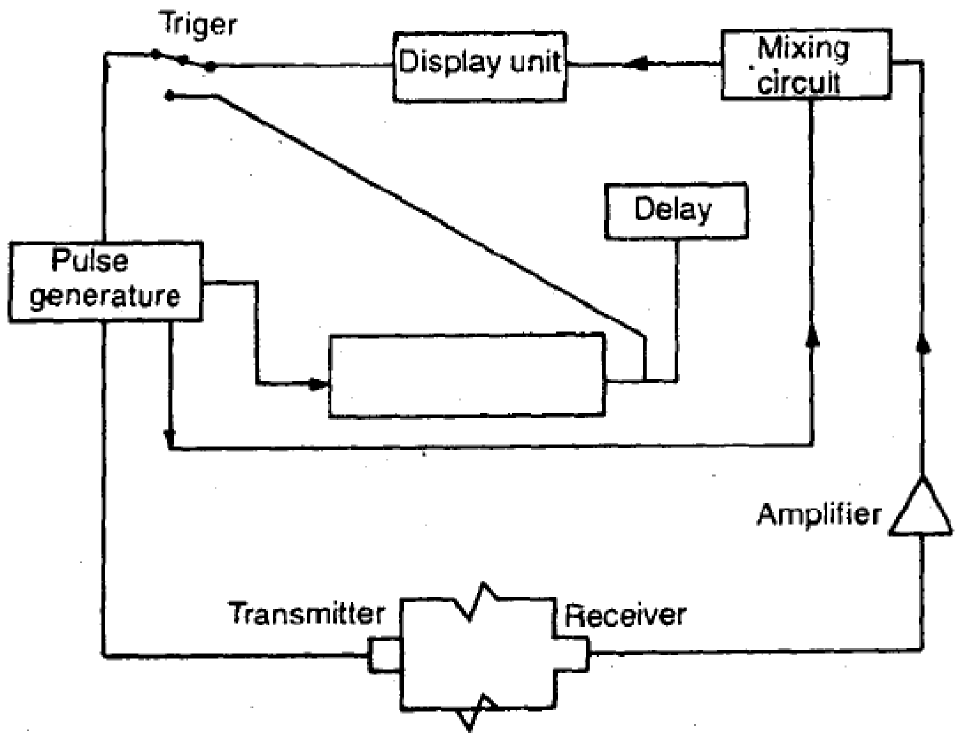


Figure 1 Typical ultrasonic pulse velocity test circuit

Note: There is no unique relation between pulse velocity and strength of concrete; however under specific conditions these two quantities are directly related. The common factor is relation of density of concrete, the change in density of concrete results in a change of velocity of pulse.

ARRANGEMENT OF TRANSDUCERS

Transducers may be arranged on concrete surface in the following three ways as shown in Fig. 2

1. On opposite face (direct transmission)
2. On adjacent face (Semi direct transmission)
3. On same face (Indirect transmission)

On opposite face

The arrangement of fixing the transducers on opposite sides has been found to be the best for the following reasons.

(a) The maximum pulse energy is transmitted at right angles to the face of the transmitter.

(b) The direct transmission method opposite side arrangement is most reliable for the measurement of transit time.

(c) The path of pulse travel is clearly defined and can be measured accurately. This approach is recommended for assessing the quality of concrete where ever possible.

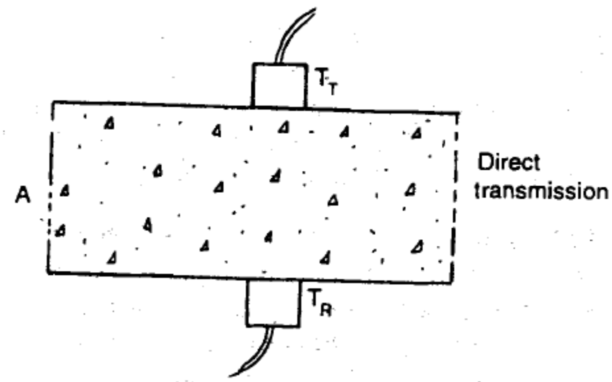


Figure 2 Typical arrangements of apparatus on opposite face

Semi direct or adjacent face method

This method can be used satisfactorily under the following conditions:

(a) If the path length is not large.

(b) The angle between the transducers is not large If these conditions are not met the clear signals will not be received due to the reduced or weak transmitted pulse.

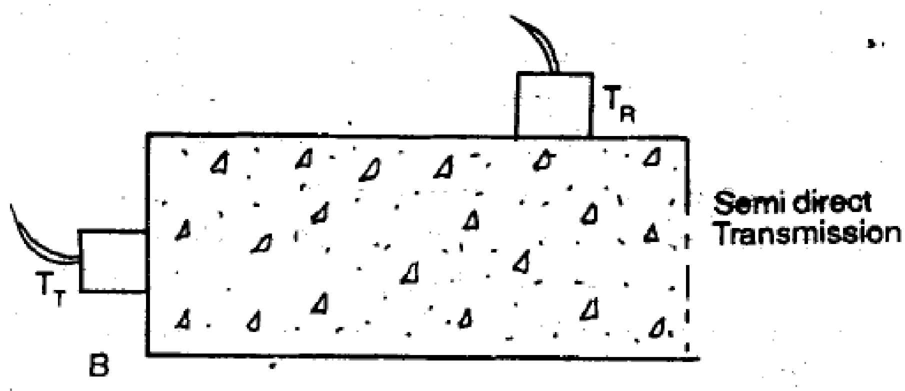


Figure 3 Typical arrangements of apparatus on adjacent face

Same face or 'indirect transmission method

In this case the pulse velocity is measured parallel to the surface of the concrete, In this case energy received is considerably low and the accuracy of readings is correspondingly decreased properties of the surface layer only are known. It does not give any indication of strength of concrete at depth. Hence it is least satisfactory. Transducers making point contact only with the surface of concrete are advantageous as no surface treatment is required. Whereas, on a flat surface, the good contact of transducers must be ensured over a considerable area. The probe transducers are more sensitive to operator pressure.

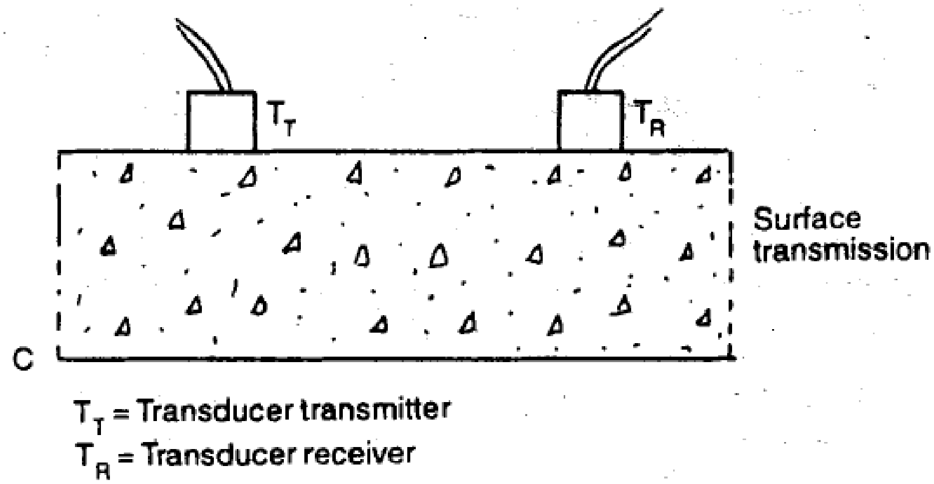


Figure 4 Typical arrangements of apparatus on same face

ADVANTAGES:

The method has provided reliable in-situ delineations of the extent and severity of cracks, areas of deterioration, and general assessments of the condition of concrete structures. The equipment can penetrate over 90 m of continuous concrete with the aid of amplifiers and is easily portable. Although most surveys are done in dry conditions, the transducers can be waterproofed for underwater surveys. Topographic inversion techniques can be used for two and three-dimensional imaging of defects by scanning the tested element with many combination of source and receiver measurement locations.

LIMITATIONS:

- Any variation of concrete temperature between 5^0C and 30^0C does not affect the pulse velocity measurement. At temperature between 30^0C and 60^0C there can be reduction in pulse velocity by 5 %. Below the Freezing temperature free water freezes with in concrete and the pulse velocity is increased by 7 %
- The pulse velocity of concrete increased with an increase in moisture content of concrete. The influence is pronounced for low-strength concrete than for high strength concrete. The pulse velocity of saturated concrete may be increased by 2%.
- The pulse velocity of steel is 1.2-1.9 times higher than the velocity through concrete which affect RCC & PCC results.

FIGURE:

PROCEDURE:

1. Preparing for use

   Before switching on the ‘V’ meter, the transducers should be connected to the sockets marked “TRAN” and ” REC”.

   The ‘V’ meter may be operated with either:

   a) the internal battery,

   b) an external battery or

   c) the A.C line.

2. Set reference

   A reference bar is provided to check the instrument zero. The pulse time for the bar is engraved on it. Apply a smear of grease to the transducer faces before placing it on the opposite ends of the bar. Adjust the ‘SET REF’ control until the reference bar transit time is obtained on the instrument read-out.

3. Range selection

   For maximum accuracy, it is recommended that the 0.1 microsecond range be selected for path length upto 400mm.

4. Pulse velocity

   Having determined the most suitable test points on the material to be tested, make careful measurement of the path length ‘L’. Apply couplant to the surfaces of the transducers and press it hard onto the surface of the material. Do not move the transducers while a reading is being taken, as this can generate noise signals and errors in measurements. Continue holding the transducers onto the surface of the material until a consistent reading appears on the display, which is the time in microsecond for the ultrasonic pulse to travel the distance 'L'. The mean value of the display readings should be taken when the unit digit hunts between two values.

Pulse velocity= (Path length/Travel time)

5. Separation of transducer leads

   It is advisable to prevent the two transducer leads from coming into close contact with each other when the transit time measurements are being taken. If this is not done, the receiver lead might pick-up unwanted signals from the transmitter lead and this would result in an incorrect display of the transit time.

UPV value in km/sec (V)	Concrete quality
V greater than 4.0	Very good
V between 3.5 and 4.0	Good, but may be porous
V between 3.0 and 3.5	Poor
V between 2.5 and 3.0	Very poor
V between 2.0 and 2.5	Very poor and low integrity
V Less than 2.0 and reading fluctuating	No integrity, large voids suspected

Interpretation of Results

Result can be interpreted in terms of quality of concrete in terms of uniformity, incidence or absence of internal flaws, cracks and segregation, etc. Indicative of the level of workmanship employed, can thus be assessed using the guidelines given below, which have been evolved for characterizing the quality of concrete in structures in terms of the ultrasonic pulse velocity.

OBSERVATIONS

Mix proportion of concrete ______________________

Size of specimen ______________________

Date of casting ______________________

Date of testing ______________________

OBSERVATION TABLE

Direct Method

Sr. No	Reading	Velocity (m/s)	Quality of concrete
1	1.		
	2.		
2	1.		
	2		
3	1.		
	2		

In Direct Method

Sr. No	Reading	Velocity (m/s)	Quality of concrete
1	1.		
	2.		
2	1.		
	2		

CONCLUSION

QUESTIONS

1. How will you classify the grade of concrete in UPV testing?

2. Enlist advantages and limitation of UPV test?
3. What properties of concretes are evaluated by UPV test?
4. State factors affecting value of UPV?

Grade Obtained :	**Sign of faculty:**	**Date:**

Concrete Technology (CL 342) **Date**

EXPERIMENT 11
COMPRESSIVE STRENGTH OF CONCRETE

OBJECTIVE

To determine compressive strength of concrete for given specimen

To develop relationships between NDT and destructive test.

APPARATUS

CTM

MATERIALS

Concrete cube / cylinder

THEORY

Compressive strength is the most important property of concrete. By definition, the ultimate compressive strength of a material is that value of uni-axial compressive stress reached when the material fails completely. The compressive strength is usually obtained experimentally by means of a compressive test. Here Digital compression testing machine is used for this experiment. Uni-axial compressive load at Uniform rate of 5.25 KN/sec is applied for 150 mm cube.

PROCEDURE

1. Take suitable quantity of ingredients for nominal mix or design concrete mix e.g 1:2:4 for nine numbers of cube specimens.
2. Mix cement and sand till uniform color is obtained. Spread this mixture over the coarse aggregate and thoroughly mix all the ingredients using a trowel.
3. Find the quantity of water required by adopting as IS method of mix design. Add the water to the mixture and mix the concrete thoroughly using the trowel.
4. Oil the inner surfaces of the steel mould. Fill the mould with concrete in 3 layers. While filling compact each layer by a standard tamping bar with 25 strokes.
5. Finish the top surface of concrete with top of the mould by means of the trowel. Then, allow the cube to dry at a temperature of 24 + 2ºC for 24 hours.

6. Remove the concrete cube from the mould and cure it under water for 7 days in a curing tank.
7. Take the concrete cube from the curing tank and remove the surface water using a cloth.
8. Measure the dimensions of the concrete cube and place it in the compression testing machine.
9. Apply the load till the cube completely crushes or fails and note down the load failure.
10. Determine the compressive strength of the given concrete cube by using the following relation:
    Compressive strength = Load failure (P) in N / Bearing area (A) in mm^2

OBSERVATION:

1. Length of the concrete cube, L = mm
2, Breadth of the concrete cube, B = mm
3. Depth of the concrete cube, D = mm
4. Load at failure, P = KN

OBSERVATION TABLE

Specimen No	Cube Failure Load	Cube Strength, N/mm^2
Day 3		
1.		
2.		
3.		
Day 7		
1.		
2.		
3.		
Day 28		
1.		
2.		
3.		

RESULT:

The Avg.compressive strength of the given concrete cube = N/mm^2

Co- relation of NDT results with compressive strength results

Grade Obtained :	Sign of faculty:	Date:

Concrete Technology (CL 342) **Date**

EXPERIMENT 12
FLEXURAL STRENGTH OF CONCRETE

OBJECTIVE

Determination of flexural strength beam specimen of concrete.

APPARATUS

CTM

MATERIALS

Beam Specimen

THEORY

The flexural strength expressed in terms of modulus of rapture and is defined as maximum tensile stress of concrete at rapture.

This is very important design parameter for concrete pavements

PROCEDURE

1. Take quantity of ingredients as per design concrete mix.
2. Mix cement and sand till uniform colour is obtained. Spread this mixture over the coarse aggregate and thoroughly mix all the ingredients using a trowel.
3. Add the water to the mixture and mix the concrete thoroughly using the trowel.
4. Oil the inner surfaces of the steel mould. Fill the mould with concrete in 3 layers. While filling compact each layer by a standard tamping bar with 25 strokes.
5. Finish the top surface of concrete with top of the mould by means of the trowel. Then, allow the cube to dry at a temperature of 24 + 2ºC for 24 hours.
6. Remove the concrete beam specimen from the mould and cure it under water for 7 days in a curing tank.
7. Take the concrete cube from the curing tank and remove the surface water using a cloth.
8. Measure the dimensions of the specimen and place it in the Flexure testing machine.
9. The load should be applied at a uniform rate and in such a manner as to avoid shocks at rate of 0.06 $\pm$ 0.04 MPa per second.
10. Apply the load till the beam completely fails and note down the load at failure.

11. Distance between line of fracture and the nearest support measured on central line of tensile side of the specimen are measured (a). If fracture occurs more than 5% outside the middle third (a<113mm) the test results are discarded.
12. Determine the flexural strength of the given concrete beam specimen by using the following relation if a>133mm.
13. $Flexural strength = \frac{M}{Z} = \frac{(W * L/6)}{bh^2/6} = \frac{Wa}{bh^2}$
14. If 113 mm < a < 133 mm flexural strength is 3 times the above equation.

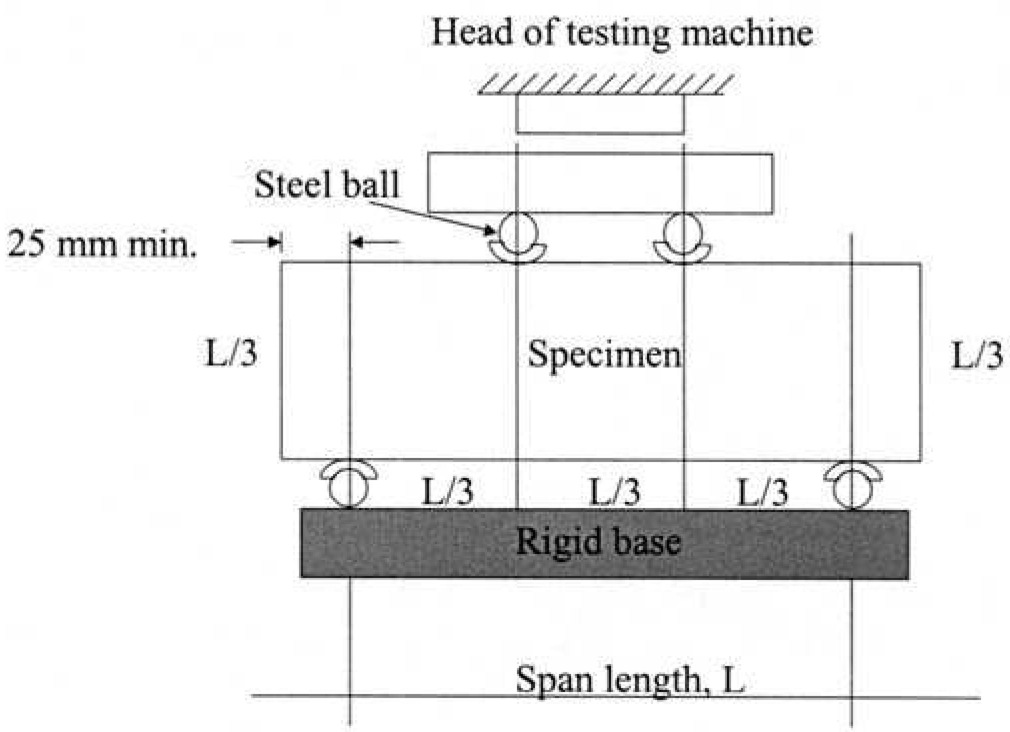


Figure 1. Flexural test arrangement

OBSERVATION:

1. Length of the concrete cube, L = mm
2. Breadth of the concrete cube, B = mm
3. Depth of the concrete cube, D = mm

OBSERVATION TABLE

Specimen No	Load at Failure, W, kN	Flexural strength MPa
Day 28		
1.		
2.		
3.		

RESULT:

CONCLUSIONS:

Grade Obtained :	Sign of faculty:	Date:

Concrete Technology (CL 342) **Date**

EXPERIMENT 13
SPLIT TENSILE STRENGTH OF CONCRETE

OBJECTIVE

Determination of split tensile strength of concrete

APPARATUS

FTM or UTM

MATERIALS

Cylinder Specimen

THEORY

The tensile strength is one of the basic and important properties of the concrete. The concrete is not usually expected to resist the direct tension because of its low tensile strength and brittle nature. However, the determination of tensile strength of concrete is necessary to determine the load at which the concrete members may crack. The cracking is a form of tension failure.

Apart from the flexure test the other methods to determine the tensile strength of concrete can be broadly classified as (a) direct methods, and (b) indirect methods. The direct method suffers from a number of difficulties related to holding the specimen properly in the testing machine without introducing stress concentration, and to the application of uniaxial tensile load which is free from eccentricity to the specimen. As the concrete is weak in tension even a small eccentricity of load will induce combined bending and axial force condition and the concrete fails at the apparent tensile stress other than the tensile strength.

As there are many difficulties associated with the direct tension test, a number of indirect methods have been developed to determine the tensile strength. In these tests in general a compressive force is applied to a concrete specimen in such a way that the specimen fails due to tensile stresses developed in the specimen. The tensile stress at which the failure occurs is termed the tensile strength of concrete.

The splitting tests are well known indirect tests used for determining the tensile strength of concrete sometimes referred to as split tensile strength of concrete. The test consists of applying a compressive line load along the opposite generators of a concrete cylinder placed with its axis horizontal between the compressive platens. Due to the compression loading a fairly uniform tensile stress is developed over nearly 2/3 of the loaded diameter as obtained from an elastic analysis. The magnitude of this tensile stress (acting in a direction perpendicular to the line of action of applied loading) is given by the formula (IS : 5816-1970):

Split tensile strength = (2*P)/(pi*L*d)

Where,
P = Load at time of failure
L = length of cylinder
D = Diameter of cylinder

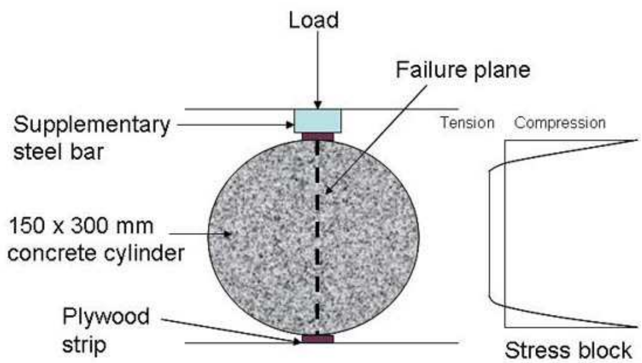


Figure 1. Test arrangement

PROCEDURE

1. Take quantity of ingredients as per design concrete mix.
2. Mix cement and sand till uniform colour is obtained. Spread this mixture over the coarse aggregate and thoroughly mix all the ingredients using a trowel.
3. Add the water to the mixture and mix the concrete thoroughly using the trowel.
4. Oil the inner surfaces of the steel mould. Fill the mould with concrete in 3 layers. While filling compact each layer by a standard tamping bar with 25 strokes.
5. Finish the top surface of concrete with top of the mould by means of the trowel. Then, allow the cube to dry at a temperature of 24 + 2ºC for 24 hours.
6. Remove the concrete cylinder specimen from the mould and cure it under water for 7 days in a curing tank.
7. Take the concrete specimen from the curing tank and remove the surface water using a cloth.
8. Measure the dimensions of the specimen and place it in the testing machine.
9. The load should be applied at a uniform rate and in such a manner as to avoid shocks at rate of 0.06 $\pm$ 0.04 MPa per second.

10. Apply the load till the specimen completely fails and note down the load at failure.

OBSERVATION:

1. Length of the concrete cylinder, L = mm
2. Diameter of Cylinder = mm

OBSERVATION TABLE

Specimen No	**Load at Failure, W, kN**	**Tensile strength MPa** (2*P)/(pi*L*d)
Day 28		
1.		
2.		
3.		

CONCLUSIONS:

Grade Obtained :	**Sign of faculty:**	**Date:**